ABSTRACT

The rapid growth in malware attacks presents a major challenge to computer systems, networks, and digital infrastructure. Traditional signature-based malware detection techniques are effective for known threats but may face difficulties in identifying previously unseen or modified malware variants. Deep learning techniques provide an alternative by automatically learning patterns from malware data. This project proposes an ensemble of lightweight Vision Transformer architectures, MobileViT and TinyViT, for malware-family classification using grayscale malware images. In image-based malware classification, malware samples are represented as images containing visual patterns associated with their byte distributions and file structures. These images are preprocessed and supplied to lightweight deep learning models for feature extraction and classification. MobileViT combines convolutional processing with Transformer-based attention to capture local and global image information. TinyViT provides another compact Vision Transformer architecture for image representation learning. The proposed ensemble combines the class-probability outputs of both models using a probability-fusion method to generate a final malware-family prediction. The project is developed using Python, PyTorch, and image-processing libraries. The Malimg malware image dataset is used to develop and evaluate the image-classification component. The current MobileViT implementation was evaluated on 957 test images belonging to 25 malware families and achieved an accuracy of 98.54%. The reported macro precision, recall, and F1-score were 0.95, 0.95, and 0.95, respectively. These results provide a baseline for the planned TinyViT and ensemble experiments. The final system will compare MobileViT, TinyViT, and the combined ensemble using accuracy, precision, recall, F1-score, parameter count, model size, and inference time. A prototype application may be developed to demonstrate static file analysis. The project also investigates the relationship between classification performance and computational efficiency. The work aims to evaluate whether an ensemble of lightweight Vision Transformers can provide effective malware-family classification while maintaining manageable computational requirements.